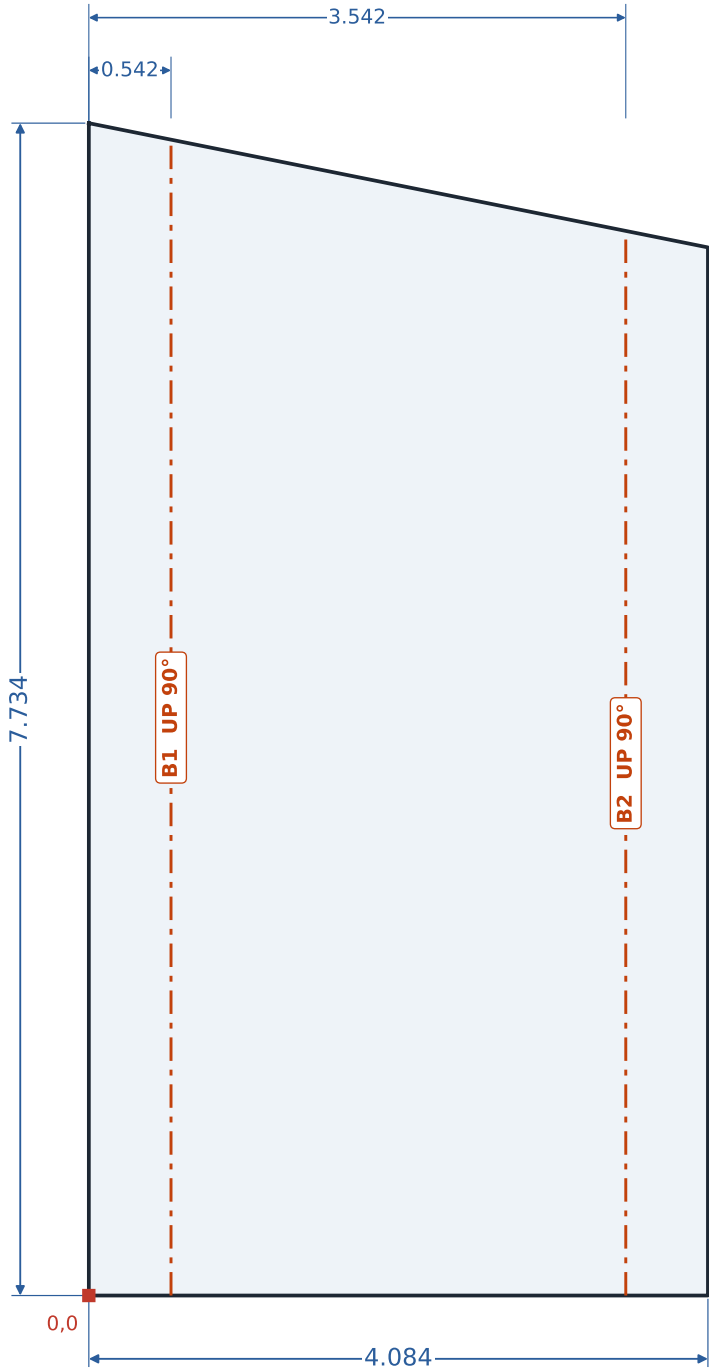


FR-3 · Channel — flat pattern (2 bends 90°)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.084" × 7.734" · ≈ 1.01 lb each · QTY 1 FR + 1 FL = 2 per vehicle

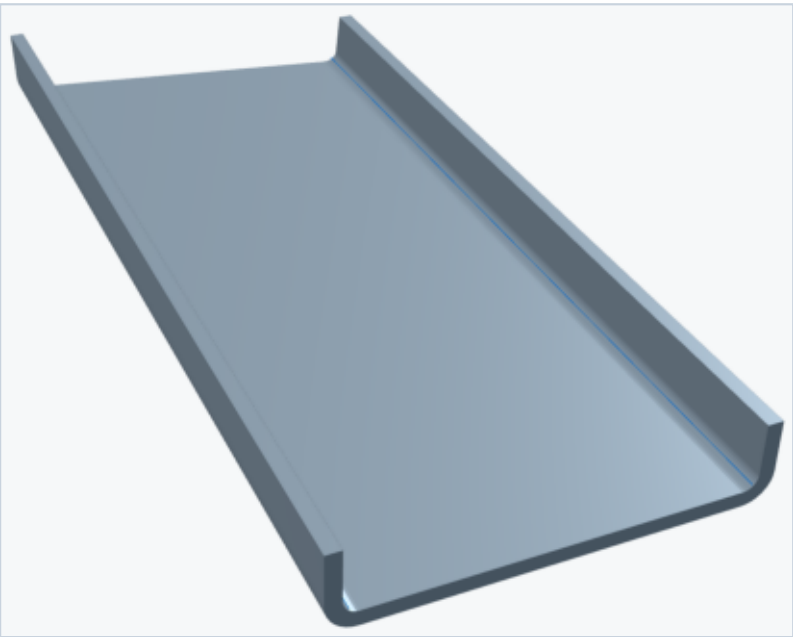
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * bend 90° on dashed lines, 3/4 flanges inward

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 0.542
B2	90.0°	UP (toward)	R=t 0.120	2	X = 3.542

FILES

- FR-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-3_ref.dxf (reference: + bend lines, drill-later)

FR-3Channel	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-3 · Channel — flat pattern (2 bends 90°)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1